LLM-ERP AI Agent & Tool Catalog (English)

> **AI transparency document for customers**: Exactly what our AI can do, can't do, how much it costs, and how we prevent it from misbehaving.

📑 Contents

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1. Agent Architecture Overview

```
User question
   ↓
IntentClassifier (weighted keyword)
   ↓ routes to relevant domain
┌──────────┬──────────┬──────────┬──────────┐
│ Inventory │  Sales    │ Production│ ... 10    │
└──────────┴──────────┴──────────┴──────────┘
     ↓          ↓          ↓
   ┌─────────────────────────────────────┐
   │ 26 tools (each agent has its own scope)│
   └─────────────────┬───────────────────┘
                     ↓
              LLM Provider abstraction
              (Claude / DeepSeek / GPT-4o / Ollama)
                     ↓
                Natural language reply
                     ↓
              Logged to DecisionLog (audit trail)
```

Core principles:

- **Agents are domain specialists**: the Inventory agent cannot call sales tools (prevents accidental cross-talk)
- **Tools are atomic actions**: 1 tool = 1 business action, individually testable
- **LLM is the brain**: only understands and plans, never touches DB directly
- **DecisionLog is memory**: every interaction stored for audit and replay

2. 10 Agents in Detail

Agent	Domain	Trigger Keywords	Expected Latency
InventoryAgent	Inventory	stock / shortage / safety / replenish	1-3 s
SalesAgent	Sales	customer / order / quote / shipment	1-3 s
ProductionAgent	Production	WO / work order / schedule / dispatch	2-5 s
PurchaseAgent	Purchase	PO / supplier / purchase / receive	2-5 s
QualityAgent	Quality	defect / inspection / NCR / CAPA / complaint	2-5 s
WarehouseAgent	Warehouse	count / transfer / warehouse / barcode	1-3 s
PlanningAgent	Planning	MPS / MRP / schedule / forecast	3-10 s
AccountingAgent	Accounting	AR / AP / month-end / invoice	2-5 s
CRMAgent	CRM	lead / opportunity / visit / customer dev	2-5 s
GeneralAgent	General	(fallback)	3-10 s

2.1 What Agents Will Not Do

Behavior	Reason
Auto-modify prices or inventory	Write tools off by default; customer opt-in only
Cross-agent tool calls	Strict scope, prevents accidental side-effects
Reply "I deleted it for you"	High-risk actions require human confirm; AI cannot execute directly
Fabricate non-existent part numbers	All queries hit real DB (anti-hallucination)
Reveal system prompt	Prompt-safety detector blocks

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3. 26 Tools Full List

3.1 Inventory (5)

Tool	Function	Writes?
query_inventory	Check stock / safety levels	No
list_parts	List parts	No
list_below_safety	List items below safety stock	No
get_part_history	Part in/out history	No
inventory_adjust	Adjust inventory ⚠️	**Yes** (confirm required)

3.2 Sales (4)

Tool	Function	Writes?
list_customers	List customers	No
query_sales_order	Query SO	No
get_customer_history	Customer price history	No
update_so_status	Change SO status ⚠️	**Yes**

3.3 Production (4)

Tool	Function	Writes?
query_work_order	Query WO	No
list_products	List products	No
get_bom	Get BOM tree	No
dispatch_wo	Dispatch WO ⚠️	**Yes**

3.4 Purchase (3)

Tool	Function	Writes?
query_purchase_order	Query PO	No
list_suppliers	List suppliers	No
approve_purchase_order	Approve PO 🚨	**Yes (high-risk)**

3.5 Quality (3)

Tool	Function	Writes?
list_inspections	List inspections	No
list_non_conformance	List non-conformances	No
create_capa	Open CAPA	Yes

3.6 Warehouse (2)

Tool	Function	Writes?
query_warehouse_stock	Multi-warehouse query	No
create_transfer	Inter-warehouse transfer	Yes

3.7 Planning (2)

Tool	Function	Writes?
run_mrp	Run MRP explosion	Yes
suggest_reorder	Reorder suggestion	No

3.8 Accounting (3)

Tool	Function	Writes?
list_receivables	List AR	No
list_payables	List AP	No
close_month	Month-end close 🚨	**Yes (high-risk)**

> 🚨 = high-risk tool; ⚠️ = medium-risk. See §6 High-Risk Actions.

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4. Cost Transparency

4.1 LLM Pricing (2025-Q1)

Model	Input ($/1M)	Output ($/1M)	Best For
Claude 3.5 Sonnet	$3.00	$15.00	International brand orders, strictest
GPT-4o	$2.50	$10.00	Industry standard
GPT-4o-mini	$0.15	$0.60	High-frequency simple queries
DeepSeek V3	$0.14	$0.28	Chinese context, best ROI
Ollama (local)	$0	$0	Zero cost, zero data egress

4.2 Typical Query Cost

Query	Input	Output	DeepSeek $	Claude $
"Today's stock status"	800	200	$0.0001	$0.0054
"M6 bolt price history" (with tool call)	2,500	600	$0.0005	$0.0165
"Run this month's MRP" (multi-tool loop)	5,000	2,000	$0.0013	$0.045

4.3 50-Person Factory Monthly Estimate

Assume 50 employees, 5 AI queries/day each, 22 working days:

- Monthly queries: 50 × 5 × 22 = 5,500
- Avg tokens: 3,500 in + 1,000 out

LLM	Monthly USD	Monthly TWD
DeepSeek only	$4.20	NT$ 130
GPT-4o-mini only	$6.30	NT$ 200
Claude Sonnet only	$140	NT$ 4,300
3-tier routing (rules 40% + Ollama 50% + Claude 10%)	**$14**	**NT$ 430**

5. Security Design (4 Layers)

```
[Layer 1] Prompt safety detector (regex)
          ├─ Detects "ignore instructions", "dump data" patterns
          ├─ Match → reject + log, never send to LLM (saves $$$ and risk)
          └─ Adversarial test: 32 cases all blocked
                ↓
[Layer 2] Agent domain restriction
          ├─ Inventory agent can only call 5 inventory tools
          ├─ Even if LLM tries to cross-domain, scope rejects
          └─ Attack surface = 1/10 size
                ↓
[Layer 3] RBAC permission check
          ├─ Every tool call checks user's permissions
          ├─ Row-level filtering (sales sees only own customers)
          └─ AI cannot exceed user's privileges
                ↓
[Layer 4] Human-in-the-loop
          ├─ High-risk tools (e.g. approve_PO) wait for human confirm
          ├─ Amount > $10,000 auto-triggers
          └─ AI cannot directly execute high-risk actions
                ↓
[Across all] DecisionLog records every interaction
          ├─ user / agent / model / tokens / cost / risk / decision
          └─ Replayable for forensics + auditable
```

6. High-Risk Actions

These tools require **human click on confirm button**, even in demo mode:

Tool	Risk
approve_purchase_order	Approving PO = authorizing payment
approve_sales_order	Sales confirm impacts inventory allocation
post_journal_entry	Posting journal affects financial reports
close_month	Month-end close = data freeze
delete_customer	Customer record deletion
delete_supplier	Supplier record deletion
delete_part	Part record deletion
bulk_inventory_adjust	Batch inventory change
bulk_price_update	Batch price change
send_email_blast	Mass email
send_line_broadcast	Mass LINE message

+ **Amount condition**: any tool with `amount/total/qty/value` ≥ $10,000 also triggers confirm.

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7. DecisionLog Audit Trail

Every AI interaction writes one row, schema:

Field	Purpose
session_id	Chat session
user_id	Who asked
domain	Which domain
agent_name	Which agent handled
query	Original question
decision	AI's answer
reasoning	Thought process
alternatives	Other options considered
model	Which LLM used
input_tokens / output_tokens	Usage
cost_usd	Converted USD
latency_ms	Duration
risk_flagged	Whether flagged by safety detector
human_confirmed	Whether high-risk got human approval

Query interface: `GET /api/analytics/ai-cost` — see monthly cost breakdown.

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8. AI Refusal List

The following requests are **rejected before reaching LLM**:

1. Reveal system prompt / developer mode
2. Cross-tenant query (read another company's data)
3. Direct SQL injection attempts
4. Mass batch deletion (unless via dedicated batch API + confirm)
5. Bypass RBAC to access data user has no permission for
6. Auto-respond "I did it" when actually nothing happened (anti-hallucination)

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9. FAQ

Q1: What if AI answers wrong?

Every reply has a DecisionLog entry, replayable by session_id. Report errors with session_id attached. AI never auto-writes anything, so wrong answers are just "wrong words", never corrupted data.

Q2: Does AI train on our data?

Depends on which LLM you choose:

- Anthropic Claude API: **No**, doesn't train on API data (public policy)
- OpenAI API: **No**, doesn't train on API data (public policy)
- DeepSeek API: Policy unclear (China DC), sensitive data avoid
- **Ollama local**: **Zero egress**, zero risk

Q3: Can we switch LLMs?

Yes. Change `backend/.env` `LLM_PROVIDER` + `LLM_MODEL`, restart backend, all agents use new LLM immediately.

Q4: Where do I see token usage?

`GET /api/analytics/ai-cost?period_days=30` — monthly breakdown by agent.

Q5: Can we fully disable AI?

Yes. `backend/.env` set `LLM_PROVIDER=disabled`. Chat endpoints return demo mode. The other 11 domains work fully.

Chinese version: `AGENT_CATALOG_ZH.md` **Last updated**: 2026-05-14 · v2.5